

TB Limiter

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Overview

Stops sharp peaks while making the overall signal louder, firmer, and more controlled.

What this plug-in solves

Final limiting needs to control inter-sample peaks without flattening musical recovery or shifting the stereo image. TB Limiter separates drive, ceiling, attack/release behavior, RMS-informed assistance, and optional balance correction so loudness and safety decisions remain explicit.

What you can expect

- Predictable true-peak ceiling control
- Stereo image stability
- Program-aware recovery and density
- Optional final 16- or 24-bit dither

How it works

TB Limiter watches incoming peaks and reduces them before they pass the chosen ceiling. This makes it possible to raise average level or protect a bus while keeping sharp transients from clipping the next stage.

Input determines how hard the limiter works, Attack and Release shape how reduction surrounds each peak, and stereo linking protects the image. Enhance and RMS assistance can change density and perceived loudness beyond simple peak catching. Add level gradually, compare at matched loudness, and leave enough output margin for the delivery format.

Quick start

- 1 Set Ceiling for the delivery target.
- 2 Raise Input Gain until the required loudness or gain reduction is reached.
- 3 Tune Release for recovery and Attack for transient shape.
- 4 Add RMS Assist if sustained material pumps or peak-only behavior feels too nervous.

- 5 Use Enhance and RMS Balance only for a specific need.
- 6 Select Dither only on the final export stage.
- 7 Compare loudness-matched and check true-peak after encoding when relevant.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Input Gain and Ceiling

Input Gain drives material into limiting. Ceiling defines the independent final maximum. Raising Input changes loudness and reduction; changing Ceiling changes delivery headroom.

Attack and Release

Attack determines how the lookahead shapes incoming peaks. Release sets the recovery speed: shorter settings feel denser, while longer settings are smoother but may reduce movement.

Enhance

Adds controlled micro-peak shaping/density. Use after basic limiting is stable because it changes transient character.

RMS Assist

Adds program-aware RMS context to recovery so sustained energy and short peaks are not treated identically.

RMS Balance

Gently corrects persistent left/right energy drift while retaining stereo-linked peak limiting. Use only when a real long-term imbalance exists.

Dither

Off, 16-bit, or 24-bit TPDF dither is intended for the final word-length reduction. Do not add dither repeatedly at intermediate stages.

Programs

Programs include Transparent, Punch, Dense Control, Balance Guard, Maximize, Mastering, and RMS Smooth starting points.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Attack	Sets how quickly the effect reacts when a sound begins. Faster values control the front; slower values keep more punch. Judge this control while the source repeats in the full arrangement. The right timing should support the groove without making the sound feel detached.
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Ceiling	Sets the highest output peak the limiter is allowed to reach. Match the final loudness before deciding whether the processing sounds better. Extra level can make almost any setting seem more impressive.
Dither	Adds the appropriate very-low-level noise when creating a 16-bit or 24-bit final file. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
Enhance	Adds controlled peak detail so the limited result feels more present and energetic. Level-match the result and listen for lost transients or harsh buildup. More color is useful only while the source still keeps its role in the mix.
Input Gain	Sets how loud the sound enters the plug-in. Raise it for a stronger response or lower it for more headroom. Match the final loudness before deciding whether the processing sounds better. Extra level can make almost any setting seem more impressive.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.
Release	Sets how quickly the effect relaxes after the sound becomes quieter. Set it in relation to the rhythm and the space between events. Listen for pumping, gaps, or a tail that covers the next sound.

Control	What it does and when to use it
RMS Assist	Adds smoother average-level control alongside the fast peak limiting. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.
RMS Balance	Corrects longer level differences between the left and right channels without chasing every peak. Move it gradually and listen for the point where the intended result becomes clear without introducing a new problem elsewhere.

Practical uses

Transparent master

- Set Ceiling with codec headroom.
- Use moderate Input Gain.
- Keep Enhance and RMS Balance low or off.
- Use medium Release and moderate RMS Assist.

Expected result: Peaks are controlled with minimal audible character.

Dense loud master

- Increase Input Gain gradually.
- Shorten Release until density rises without obvious chatter.
- Add Enhance carefully.
- Recheck low-frequency distortion and stereo stability.

Expected result: The master becomes louder and denser while remaining under the chosen ceiling.

Common pitfalls

A limiter cannot repair clipping already printed into the source. Enable only the repair section that matches the problem and use the smallest amount that makes the noise less distracting without removing useful detail.

True-peak compliance should be rechecked after lossy encoding. Reduce the main amount, feedback, drive, or input first, then rebuild the sound while watching the output meter and listening after the source stops.

Dither belongs at the last bit-depth reduction only. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.